National University of Computer and Emerging Sciences

Information Retrieval (CS4051)

Date: February 24, 2025

Course Instructor

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Sessional-I Exam

Total Time: 1 Hour

Total Marks: 40

Total Questions: **03**

Semester: Spring-2025

Campus: Karachi

Department: AI & DS

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CLO # 1: < Understand the basic concepts and techniques in Information Retrieval.>
CLO # 4 < Appreciate the importance of data structures, indexes and retrieval efficiency>

Q1. Answer the following questions to the point. **[Time: 30 mins] [Marks: 2x10]**

a. What are stop-words in the context of IR? What type of query requires us to keep the stop-words in indexing?

In Information Retrieval (IR), stop-words are commonly occurring words that are usually ignored during indexing and retrieval because they provide little meaningful information for search queries. Removing stop words save some storage cost but we cannot be able to answer phrase queries or Named Entity searches. E.g "to be or not to be" or "prince of persia"

b. What are some of the assumptions we did before building a Boolean Retrieval Model?

Documents are composing of features. We are able to process these documents automatically through parsing algorithm.
Users are aware of these features in the documents and they come up with the queries following these features.

c. Illustrate one situation in which we would need tolerant retrieval?

A situation in which user is not sure about the actual spelling of a term she is looking for, she can run the wildcard query like “rel*ion”. This ensures better user experience and higher recall in search applications.

d. What are context sensitive spelling corrections? Give an example.

Context-sensitive spelling correction refers to a method of correcting spelling errors by considering the surrounding words in the sentence. Unlike basic spell checkers that correct isolated misspellings, this approach ensures that the correction makes sense within the context of the sentence. for example, "I no their going to the party." likely to be fixed as "I know they’re going to the party."

e. Does Lemmatization Improve Recall in Information Retrieval? Give justification.

Yes, lemmatization can improve recall in Information Retrieval (IR), Lemmatization reduces words to their base (lemma) form, allowing retrieval of different inflected forms of a word and handles morphological variations. It also reduces the vocabulary size for inverted index.

f. What is the Connection Between Heap’s Law and Zipf’s Law?

Heap's Law and Zipf's Law are complementary; Zipf's Law explains the frequency distribution of words. While, Heap's Law explains how the vocabulary size grows as the text size increases, which is influenced by the frequency distribution described by Zipf's Law. Both laws are fundamental to understanding the statistical properties of natural language and are widely used in information retrieval to estimate inverted index size for a given collection.

g. Define Context-Sensitive Spelling Correction, Give an example query

Context-sensitive spelling correction refers to a method of correcting spelling errors by considering the surrounding words in the sentence. Unlike basic spell checkers that correct isolated misspellings, this approach ensures that the correction makes sense within the context of the sentence. for example, "flight form China." likely to be fixed as "flight from China."

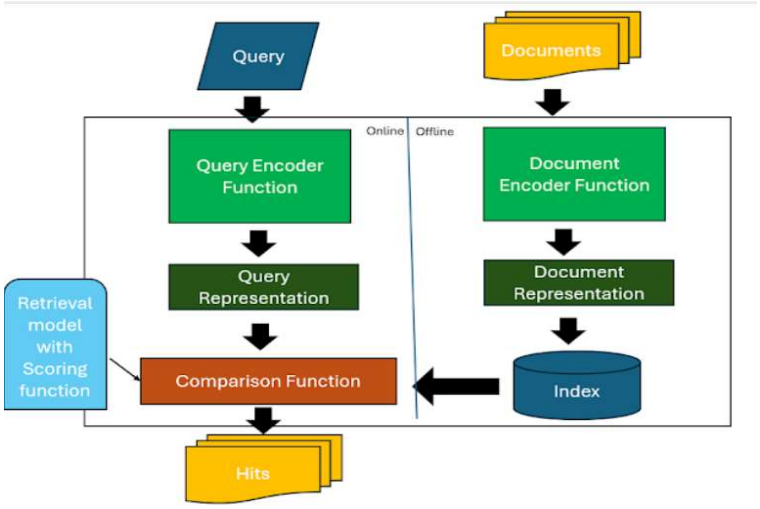
h. What are the shortcoming of Block-Sort Based Indexing (BSBI) approach?

Block-Sort Based Indexing (BSBI) is a disk-based indexing algorithm used for constructing an inverted index when dealing with large datasets. It processes documents in blocks, sorts term-posting pairs, and then merges them. However, despite its efficiency, BSBI has some limitations:(1) High Memory Usage per Block and (ii) Sorting Overhead.

i. Outline two major limitations of Vector Space Model (VSM) for IR.

VSM lost the original word order from the document and this can be very important information for search.
VSM does not account synonyms into the model and hence smart and bright are independent. This is termed as vocabulary mismatch problem.

j. Illustrate with the help of a diagram- the entire process of Information Retrieval from a given text query.



CLO # 2: <Understanding how IR Model works >

Q2. Consider the following documents in a collection. **[Time: 15 mins] [Marks: 2x5]**

Doc 1: w1 w3 w5 w3
Doc 2: w3 w2 w7
Doc 3: w2 w4 w1
Doc 4: w4 w2 w6 w3

a. Identify the vocabulary of the collection. What is the size of the vocabulary? [2]

V = {w1, w2, w3, w4, w5, w6, w7} and |V| = 7

b. Give the document vector for Doc 4, using on term frequency. [2]

d_4 = < 0, 1, 1, 1, 0, 1, 0>

c. If the query= w1 w2 what will be the cos(angle) between Doc1 and query, using term frequencies as coefficient of the vectors? [2]

q = < 1, 1, 0, 0, 0, 0, 0> => |q| = √2
d_1 = < 1, 0, 2, 0, 1, 0, 0> => | d_1 | = √6
q dot d_1 = < 1, 1, 0, 0, 0, 0, 0> . < 1, 0, 2, 0, 1, 0, 0> = 1
Hence Cos (d_1, q) = 1/ √12

d. Find the Projection of Doc 2 vector on Doc 4 vector? [2]

d2 = <0, 1, 1, 0, 0, 0, 1>
d4 =<0, 1, 1, 1, 0, 1, 0>

Proj d2 on d4 = {(d_2 * d_4) / (d_4 . d_4) } X d_4 = 2/4 X <0, 1, 1, 1, 0, 1, 0>
Proj d2 on d4 = <0, 1/2, 1/2, 1/2, 0, 1/2, 0>

e. Find the cos(angle) between Doc 2 and Doc 3. [2]

d_2 = < 0, 1, 1, 0, 0, 0,1> => | d_2| = √3
d_3 = < 1, 1, 0, 1, 0, 0, 0> => | d_3| = √3
d_2 dot d_3 = < 0, 1, 1, 0, 0, 0,1>. < 1, 1, 0, 1, 0, 0, 0>= 1
Hence Cos (d_2, d_3) = 1/ 3

CLO # 3: < Appreciate the importance of data structures, indexes and retrieval efficiency >

Q3. Answer the following questions. **[Time: 15 mins] [Marks: 2x5]**

a. Construct a bi-gram index for the terms {elite, elope, ellipse}

Bi-gram	Terms
$e	elite, elope, ellipse
e$	elite, elope, ellipse
el	elite, elope, ellipse
ip	ellipse
it	elite
li	elite, ellipse
ll	ellipse
lo	elope
op	elope
pe	elope
ps	ellipse
se	ellipse
te	elite

b. What will be the bi-gram index term, when we try to fetch index for the query "rel*ion"? give one true positive and one false negative words for the given wild card query?

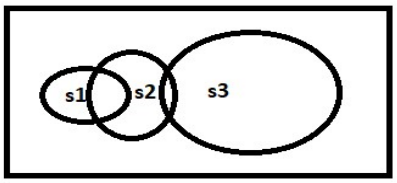
The bi-gram inex terms for term rel*ion will be "rel*ion" → { "re", "el", "io", "on" }, a true positive word will be "religion" and a false negative word will be "rellion"

c. Suggest some approach that can be used to compressed the dictionary from an IR prospective?

Dictionary-as-a-String (Blocking) -Instead of storing individual words separately, store the dictionary as a single concatenated string is more efficient in term of space. WE need to maintain an offset table to indicate where each word starts. This also reduce the pointer overhead. By this approach we can save 20% storage costs on average case.

d. For a conjunctive query, is processing postings lists in order of size guaranteed to be optimal? Explain why it is, or give an example where it isn't.

The order is not guaranteed to be optimal. Consider three terms with postings list sizes s1 = 100, s2 = 105 and s3 = 110. Suppose the intersection of s1 and s2 has length 100 and the intersection of s1 and s3 length 0. The ordering s1, s2, s3 requires 100+105+100+110=315 steps through the postings lists. The ordering s1, s3, s2 requires 100+110+0+0=210 steps through the postings lists.



e. What is the effect of skip length on the intersect algorithm? How can we determine the effective size of the skip pointer?

If the skip length is large, the algorithm can skip more elements at a time, reducing the number of comparisons. On the other hand, If the skip length is too small, the algorithm performs frequent skips but with smaller jumps, leading to more pointer adjustments. The optimal skip length depends on balancing skipping efficiency and overhead. A common heuristic is:
Skip Interval $=\sqrt{P}$, where P is the size of the posting list for term t.